Jashore University of Science and Technology

Jashore-7408, Bangladesh

EV Charging Load Demand Prediction Using Machine Learning in a SUMO-Based Simulation Environment

Submitted By:

Rakesh Biswas

Student ID: 200112

Registration Year: 2022

Academic Year: 2020–2021

Supervisor:

Monishanker Halder

Assistant Professor, Department of CSE

Jashore University of Science and Technology

This thesis report was submitted in partial fulfillment of the

Requirements for the Bachelor's Degree

In **Computer Science and Engineering**

Faculty of Computer Science and Engineering

April 2026

Jashore University of Science and Technology

Jashore-7408, Bangladesh

This thesis entitled **"EV Charging Load Demand Prediction Using Machine Learning in a SUMO-Based Simulation Environment"** submitted by **Rakesh Biswas**, Student ID: 200112, has been accepted as satisfactory in fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering.

Signature of the Student

Rakesh Biswas

Signature of the Supervisor

Dedication

To my beloved parents, whose endless love, sacrifice, and encouragement have been the cornerstone of every achievement in my life.

To my teachers, who ignited the flame of curiosity within me and guided me towards the pursuit of knowledge.

And to all the researchers and engineers working tirelessly towards a sustainable, electrified future of transportation.

Acknowledgments

First and foremost, I would like to express my deepest gratitude and sincere thanks to my supervisor, **Monishanker Halder**, for his invaluable guidance, constant encouragement, and expert advice throughout this research. His/her constructive feedback and unwavering support were pivotal in shaping this work.

I extend my appreciation to the faculty members of the Department of Computer Science and Engineering (CSE) at Jashore University of Science and Technology (JUST) for providing an intellectually stimulating environment and for their continued support during my graduate studies.

Special thanks are due to my fellow graduate students and labmates for their camaraderie and many productive conversations. I also wish to acknowledge the open-source communities behind Simulation of Urban Mobility (SUMO), Python, scikit-learn, and XGBoost, whose tools formed the backbone of this research.

Finally, I owe immeasurable gratitude to my family for their patience, prayers, and moral support. This work would not have been possible without them.

Rakesh Biswas

Department of CSE, JUST

April 2026

Table of Contents

List of Figures

List of Tables

Abstract

The rapid growth of Electric Vehicle (EV)s presents both an important opportunity and considerable challenge for modern power grid management. To optimize energy distribution, accurate prediction of EV charging load demand is critical. It reduces peak-hour stress on electrical infrastructure and enables smart grid operations. This thesis presents a comprehensive framework for predicting EV charging load demand using supervised Machine Learning (ML) techniques applied to data generated from a realistic microscopic traffic simulation. To enable real-time vehicle monitoring and intelligent charging station management, the simulation environment was built by integrating the open-source urban mobility simulator SUMO with the Traffic Control Interface (TraCI). A network of 10 bidirectional charging stations was deployed with configurable power ratings ranging from 25 kW to 50 kW. Also, separate charging and waiting queue slots were configured for each station. Forty-nine electric vehicles of the battery-powered type were simulated for 600 seconds. This resulted in 34,020 vehicle-level records, including features such as state of charge (SOC), speed, acceleration, station utilization, queue length, and system-wide load metrics. Several supervised ML models were trained and evaluated for system-level load demand forecasting: Decision Tree, Random Forest, XGBoost, and Linear Regression. The performances of the models were evaluated using mean absolute error, root mean square error, and coefficient of determination (R^2) scores. Both Decision Tree and Random Forest achieved perfect scores, whereas XGBoost achieved a near-perfect result. Linear Regression also showed strong performance. In addition, a rule-based smart charging recommendation system was implemented. In this scenario, vehicles were routed to the optimal charging station based on real-time SOC level, distance to the nearest station, slot availability, and queue status. The results show that SUMO-based simulation is an effective and efficient method for generating enriched EV charging datasets, and tree-based ensemble methods are particularly suitable for short-term load demand forecasting in EV charging infrastructure.

Keywords: Electric Vehicle, Charging Load Prediction, Machine Learning, SUMO Simulation,

Random Forest, XGBoost, Smart Charging, SOC, Queue Management.

Chapter 1

Introduction

Chapter 1: Introduction

The global transportation sector is undergoing a radical change due to the urgent need to reduce greenhouse gas emissions and to reduce dependence on fossil fuels. For totally decarbonizing road transport, Electric Vehicle (EV)s have emerged as one of the most promising technologies. According to the [1], the number of electric vehicles worldwide exceeded 26 million in 2022 and is forecast to continue to grow rapidly in the coming decades. With the rapid growth of electric vehicle usage, the increasing demand for electricity for charging is becoming a serious concern for both power system managers and urban planners [2].

With the rapid growth of EVs, the need for intelligent control infrastructure for energy stability and efficiency of optimal use has become significant. Three-wheeled electric vehicles such as Easy Bikes, Electric Rickshaws, and Electric Vans in developing countries like Bangladesh have accelerated the need for scalable charging solutions [3]. These vehicles play a major role in urban and semi-urban mobility. However, their unbalanced charging behavior can lead to sudden fluctuations in electricity demand, which creates pressure on the power grid.

EVs communicate directly with the power grid. Uncoordinated charging behavior—especially during peak hours—can create significant stress on the distribution network. This can lead to voltage violations, transformer overloads, and increased energy costs [4]. Therefore, effective management of EV charging loads requires accurate forecasting tools capable of estimating spatiotemporal demand patterns across the charging station network [5].

In this field, one of the key challenges is to accurately predict system-level charging load demand. Reliable load prediction enables better planning of charging stations, improved power distribution, and the prevention of overload situations.

1.1 Background

The rapid increasing of electric vehicles is reshaping transportation and electricity distribution systems. Unlike conventional fueled vehicles, electric vehicles, shift energy demand from fuel stations to the power grid . This creates loads that vary over time and space. Understanding the movement, energy consumption and charging behavior of electric vehicles is essential for load prediction, grid reliability and optimal utilization of charging infrastructure. In cities of Bangladesh , a major drawback is the unorganized and untested charging behavior of electric vehicles . Vehicles oftenly crowd around specific nodes or roadside charging points. Which is resulting in large amounts of energy consumption during peak hours . This often results in stress on the local grid, voltage drops and overloading of transformers problems. Which that already exist in rapidly growing cities like Khulna, Jashore, Satkhira. Besides, energy consumption, charging-load varies greatly depending on vehicle speed, congestion levels, route selection, and driver behavior . Without proper modeling and prediction of these parameters, city authorities and electricity distributors face increasing difficulties in ensuring grid reliability, planning future charging stations and controlling unbalanced load distribution. This research work uses traffic simulation tools such as SUMO to model real-world urban mobility. Where EV extensions to SUMO help model battery behavior, charging events, energy consumption, system load behavior. These outputs are integrated with load prediction models to create a robust approach for analyzing the load impact of EV adoption.

1.1.1 Definition

1.1.1.1 Battery-Powered Vehicle (BPV) or EV

A Battery Powered Vehicle(BPV) or Electric Vehicle (EV) is a transportation system which is powered entirely by rechargeable batteries and electric motors instead of an internal combustion engine. EVs store electrical energy in lithium-ion or lead-acid batteries and through an electric drivetrain, they convert this stored energy into mechanical energy . In the context of Bangladesh in urban areas like Khulna and Jessore -electric vehicles include the widely used low-cost electric three-wheelers (e.g., easy bikes , e-rickshaws, e-vans). Those vehicles are popular due to their low operating costs, absence of exhaust emissions and suitability for short-distance travel within cities or urban area . Their electricity-based operations directly link transportation demand to local electrical grid-loads. Which is making EV behavior a critical element in grid planning.

1.1.1.2 Load Prediction

Load prediction refers to the process of estimating the total electrical demand of a system based on historical or observed data. In this study, it specifically refers to the prediction of the system-level charging load (`system_total_load_kW`) generated by electric vehicles using machine learning models . The prediction is performed by understanding the relationship between various input characteristics —such as time progress , number of vehicles charging, system-level vehicle statistics and battery charge status and the corresponding load demand . Unlike traditional analytical approaches , this research uses data-driven models to understand the complex and nonlinear dependencies within the system. Proper load prediction can help to understand in better way how electric vehicle charging behavior affects overall energy demand and helps in efficient planning and management of charging infrastructure.

1.1.1.3 Traffic Microsimulation

Traffic microsimulation is a precise, vehicle-level modeling technique that which captures the detailed movements of individual vehicles over time. It represents the interaction of speed, acceleration, deceleration, lane change behavior, path selection, stop-and-go cycles and congestion with high temporal accuracy. Microsimulation becomes increasingly important in EV research. As vehicle speed directly affects the dynamics of energy consumption and battery degradation. By integrating microsimulation with EV models , researchers can generate realistic estimates of power consumption, load behavior under different road, time and traffic conditions. Which ultimately helps in more accurate charging load analysis and prediction .

1.1.1.4 State of Charge (SOC)

State of Charge (SOC) is a quantitative measurement which indicates the remaining energy stored in an electric vehicle (EV) battery. It is usually expressed as a percentage (0–100%). SOC decreases when the vehicle is being driven due to the use of propulsion power and it increases when the vehicle is connected to a charging station. As it determines when and where a vehicle will need charging , SOC is a fundamental variable for understanding electric vehicle behavior . In simulation environments like SUMO, SOC-logs give : time-step-level insights into real-time power consumption. This enables the creation of detailed charging demand models and node-level load prediction profiles.

1.1.1.5 Charging Station

A charging station is a road-side power supply point where electric vehicles (EVs) recharge their batteries. In simulation-based studies, charging stations are modeled as infrastructure elements . Those are connected to specific network nodes or road segments. Each station has certain characteristics, such as : charging capacity, availability, deployment strategy and charging session duration.

1.1.1.6 Simulation of Urban Mobility (SUMO)

SUMO (Simulation of Urban Mobility) is an open-source, microscopic traffic simulation platform . It is used for modeling and analyzing transportation networks. SUMO supports user-defined road networks, custom vehicle flows, routing algorithms, and mobility patterns. Through its EV extension, Sumo can simulate battery discharge, energy consumption, regenerative braking and real-time charging events. Sumo provides a controlled environment for research on the EV ecosystem in Bangladesh. Here in SUMO urban traffic behavior can be simulated and the resulting charging demand can be measured without the need for large-scale real-world experiments.

1.1.1.7 Traffic Control Interface (TraCI)

The Traffic Control Interface (TraCI) is SUMO's external programming interface . It allows real-time interaction between the simulation and external applications. TraCI enables dynamic control of vehicle characteristics, simulation variable extraction, traffic signal management and continuous logging of EV parameters such as SOC, speed, energy consumption and charging events. In EV load prediction research, TraCI plays an important role in automated data collection . It also integrates SUMO outputs with Python-based analytical models for prediction and visualization.

1.1.1.8 Power Consumption Analysis

Power consumption analysis refers to the study of how electric vehicles (EVs) use electricity across different driving conditions, speeds, accelerations and network paths. This includes quantifying energy consumption per route, per node and per unit distance . Which helps researchers to understand which locations or routes are putting the most pressure on the grid. In urban environments like Khulna and Jessore, this type of analysis is really significant for identifying charging hotspots, peak demand periods and areas that need infrastructure reinforcement.

1.1.1.9 Load Forecasting Model

A load forecasting model is a computational method—often statistical or machine learning-based. It is used to predict future EV charging loads using past charging shortages, route flows , traffic congestion and charging station usage patterns. These models help predict upcoming power demand, optimize the placement of charging stations. They also guide power suppliers in planning feeder capacity expansion. With the rapidly expanding use of electric vehicles in Bangladesh, such forecasts are essential to prevent transformer overloading and grid instability.

1.1.1.10 Urban EV Mobility Pattern

Urban electric vehicle mobility patterns refer to the characteristic movement behavior of electric vehicles within a city. It includes typical routes, peak congestion times, center-level activity , trip length and charging intervals. By understanding mobility patterns, researchers can link electric load distribution to traffic dynamics . In Bangladeshi cities —where electric vehicles like Easy Bikes, E-Rickshaws follow regular routes between markets, hospitals and government institutions. This analyzing mobility patterns becomes essential for accurate simulation modeling and load forecasting.

1.1.1.11 Queue Management and Promotion Logic

Queue management is an infrastructure-level process -which coordinates the flow of vehicles at a charging station. It distinguishes between active charging slots (where energy is transferred) and waiting slots (where vehicles are queuing for service). Promotion logic refers to the systematic process of moving a vehicle from a waiting state to a charging state as soon as a slot becomes available. This typically follows a first-in, first-out (FIFO) protocol to ensure fairness and simulation accuracy .

1.1.1.12 Rule-Based Recommendation System

A rule-based recommendation system is a decision-making algorithm -which uses some predefined "if-then" conditional statements to suggest the best optimal actions . In the case of EV charging, this system evaluates real-time telemetry. Such as a vehicle's SOC, distance to the nearest station and current queue length to recommend the most suitable charging locations . It also balances the overall load of the network with the needs of each vehicle.

1.1.1.13 Heterogeneous EV Fleet

A heterogeneous EV fleet refers to a collection of different types of electric vehicles within a single simulation or study. It is characterized by different physical sizes, battery capacities and power requirements. In urban modeling in Bangladesh, this primarily includes a mix of easy bikes, electric rickshaws and electric vans . It is essential to take this diversity into account to capture the complex and uneven energy usage patterns. This is typical of real-world traffic.

1.1.1.14 Spatiotemporal Demand Pattern

Spatiotemporal demand pattern refers to the distribution of electrical load, across both space (geographical location of charging nodes) and time (simulation period or daily interval) .

By analyzing these patterns, researchers and service providers can identify charging 'hotspots' and peak demand times. Wchich, enabling more precise grid reinforcement and infrastructure planning.

1.2 Problem Statement

The fast increasing of electric vehicles (EVs), specially in urban transportation systems, has created significant challenges in accurately prediction the charging load demand on the electricity grid. In Bangladesh, where electric three-wheelers such as e-bikes and electric rickshaws are widely used [3], the electricity distribution infrastructure is often already under stress during peak hours. The addition of EV charging intensifies this challenge due to the highly dynamic and nonlinear charging behavior . Which is influenced by factors such as -vehicle movement, battery state of charge , charging station availability and vehicle distribution at the system-level. A major limitation of existing research is the lackings of comprehensive and large scale real world datasets . That capture both vehicle and station level features, especially in developing areas. Besides, many conventional and statistical methods fail to effectively model the complex relationships between these variables. Where some advanced models can be computationally expensive and less intuitive for practical application in resource-limited environments . So, there is a critical need for a data-driven, effective and understandable approach to system level EV charging load demand prediction. This study addresses those challenges by using a simulation-based dataset that includes detailed system and vehicle characteristics . It implementes and evaluates multiple machine learning models to accurately predict the overall load demand (*total system load kW*) under changing system conditions.

1.3 Objectives

The primary objectives of this research are follows :

1. To design and implement a SUMO-based EV traffic simulation with realistic charging

station infrastructure, queue management and smart rerouting logic.

2. To generate a comprehensive, ML ready dataset, capturing per-vehicle and system-level features across a 600-second simulation run.

3. To train and evaluate four supervised ML models —Linear Regression (LR), Decision Tree (DT), Random Forest (RF), Extreme Gradient Boosting (XGB) —for predicting the system-level EV charging load demand.

4. To implement a rule-based charging recommendation system that suggests optimal charging stations to vehicles based on proximity, slot availability and priority.

5. To analyze model performance using standard regression metrics (Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Coefficient of Determination (R^2)) and identify the most suitable approach for this domain.

1.4 Research Questions

This study seeks to answer the following research questions :

1. Can a SUMO-based microscopic simulation environment generate a comprehensive and realistic dataset for modeling EV charging load demand?

2. Which machine learning model provides the most accurate prediction of system-level EV charging load (`system_total_load_kW`)?

3. What are the most influential features affecting EV charging load demand in a multi-vehicle system?

4. How effectively can machine learning models capture the relationship between system-level and vehicle-level features for accurate load prediction?

5. How does the performance of linear models compare with tree-based models for EV load prediction?

6. How accurately can the trained models predict load demand across different system conditions?

1.5 Significance of the Study

The significance of this research lies in contributing to the growing challenge of managing electric vehicle (EV) charging demand through a data driven and scalable approach. With the rapid growth of electric vehicles, specially in developing regions like Bangladesh [3] —power distribution systems are increasingly facing unpredictable and highly variable load patterns due to unbalanced charging behavior. This study provides a realistic framework for accurately predicting system-level EV charging load (`system_total_load_kW`) using a comprehensive simulation-based dataset . It incorporates both vehicle-level dynamics and system-level operational characteristics. By applying and comparing multiple machine learning models, this study shows that advanced yet in computationally inexpensive techniques, particularly tree-based models- can effectively capture the complex nonlinear relationships inherent in the EV charging ecosystem. The study results highlight that load demand is largely driven by a few key system-level factors. Which is enabling stakeholders to better understand the underlying determinants of energy consumption. It has a direct impact on grid management, which includes improved demand prediction, optimal power distribution, improved planning of charging infrastructure and peak load stress mitigation on transformers and substations. Moreover, the use of understandable models and feature importance analysis ensures that the results are not only accurate but also interpretable. This is making them suitable for application in resource limited environments. Overall , this research bridges the gap between simulation-based data generation and real-world applicability. And this is providing a strong foundation for future research and practical applications in intelligent energy management systems for EV-integrated smart grids.

Chapter 2

Literature Review

Chapter 2: Literature Review

The research field related to EV charging load prediction and smart charging management encompasses multiple disciplines, including power systems engineering, transportation science and computer science . This chapter reviews the most relevant works in three thematic areas: (i) EV charging load modeling and pediction, (ii) simulation-based methods for EV data generation and (iii) machine learning methods for energy load forecasting.

2.1 EV Charging Load Modeling

Early approaches to EV charging load modeling dependent on probabilistic approaches, which used statistical distributions of arrival times, departure times and daily travel distances to generate artificial load profiles [6]. Sortomme et al. [4] showed that integrated charging strategies can significantly reduce distribution system losses compared to unregulated charging. He et al. [7] discussed the optimal placement of charging stations. Which showed that accessibility and driving range limitations play an important role in station usage patterns. More recent researches have adopted a data-driven approach. Chen et al. [8] integrated probabilistic load flow analysis with EV charging uncertainty to assess grid impact. This provides a basis for the feature engineering techniques. Which is also used in the present work. Ul-Haq et al. [6] modeled residential EV charging patterns and highlighted the importance of capturing spatial heterogeneity across charging locations.

2.2 Machine Learning for Load Forecasting

ML has become the dominant paradigm for short-term electricity load prediction. Zhang et al. [5] proposed a comprehensive deep learning framework for EV charging load forecasting. Which incorporated historical demand, time-of-day characteristics and weather data, achieving significant improvements over classical autoregressive models. Wang et al. [9] applied random forest (RF) regression in short term load forecasting at EV charging stations. That is demonstrating robustness against noisy sensor data and missing values. Gradient boosting methods have also achieved state-of-the-art results. Chen et al. [10] applied XGBoost to one-day EV load forecasting That is exploiting its regularization ability to prevent overfitting on small datasets. Tayeb and Hyndman [11] compared gradient boosting with conventional time series methods. And they found that tree-based methods consistently outperformed ARIMA models for nonlinear load patterns. Although Linear regression, simple remains a useful basis for load forecasting. Guo et al. [12] showed that linear models can capture the dominant trends in EV charging demand when the features are carefully designed, especially when polynomial and interaction terms are included.

2.3 SUMO-Based EV Simulation

SUMO has been widely adopted for EV research because of its open-source nature, extensibility and support for battery models. Lopez et al. [13] describe the core capabilities of SUMO, including models of electric vehicle energy consumption, parking areas and a Traffic Control Interface (TraCI) API for external control. Wegener et al. [14] introduced the TraCI interface, that enables real time monitoring and manipulation of simulated vehicles. This makes it ideal for implementing adaptive charging management algorithms . Helbing and Triber [15] established the theoretical foundations for microscopic car-following models. That are the basis of SUMO's vehicle dynamics. The integration of these dynamics with electric powertrain models allows for accurate simulation of energy consumption patterns. And this directly affects the state of charge (SOC) trajectory and charging events.

2.4 Smart Charging and Recommendation Systems

Yilmaz and Crain [2] gave a comprehensive review of vehicle-to-grid technologies, identifying intelligent charging coordination as a key enabler of grid stability . Ding et al. [16] proposed a two-stage robust optimization for distributed energy management. That concept is related to the queue-aware routing logic implemented in this work. For EV charging coordination, Nguyen et al. [17] developed an adaptive dynamic programming method . Which showed that real-time SOC aware routing can significantly reduce the average waiting time. This motivates the rule based recommendation engine developed in the present study. Which selects stations based on SOC priority level and slot availability.

2.5 Research Gaps

Despite significant work in this area, several gaps remain. Most existing research relies on real-world datasets that are difficult to obtain, especially in the context of developing countries [3]. The creation of simulation based datasets using SUMO with integrated queue management and real-time recommendation systems has not been widely explored. Moreover, comparative benchmarking of multiple ML models on simulated EV load data is limited in the literature . This thesis addresses these shortcomings through an integrated simulation, data generation and ML prediction framework.

Chapter 3

Methodology

Chapter 3: Methodology

This chapter describes the comprehensive research methodology developed for predicting EV charging load demand. The approach integrates microscopic traffic simulation with advanced machine learning techniques to capture the complex, stochastically driven dynamics of urban energy consumption. The methodology is structured to provide a replicable framework for evaluating charging infrastructure impacts in emerging urban centers.

3.1 Research Design

This research follows a quantitative, simulation-based experimental design centered on the "Monitor–Predict–Control" paradigm. Which is designed to bridge the gap between microscopic dynamics patterns and grid level electrical load - demand. As shown in Figure 3.1, the research design is performed through a systematic multi stage workflow. It begins with extensive **Data Collection** and **Environment Setup** on the simulation platform. Using this foundation, a **Custom Road Network Design** was performed to model the real characteristics of urban area of Bangladesh [3] (as for sample Khulna city's particular are is selected). Then **Traffic Light System Placement** and **Charging Station Placement** were strategically placed to simulate realistic infrastructural limitation and constraints. After the establishment of the physical environment , the approach shifts to modeling by determining **Vehicle Configuration in SUMO** [13] and **Vehicle Fleet Characterization**, representing the operational behavior of electric three-wheeled vehicles.

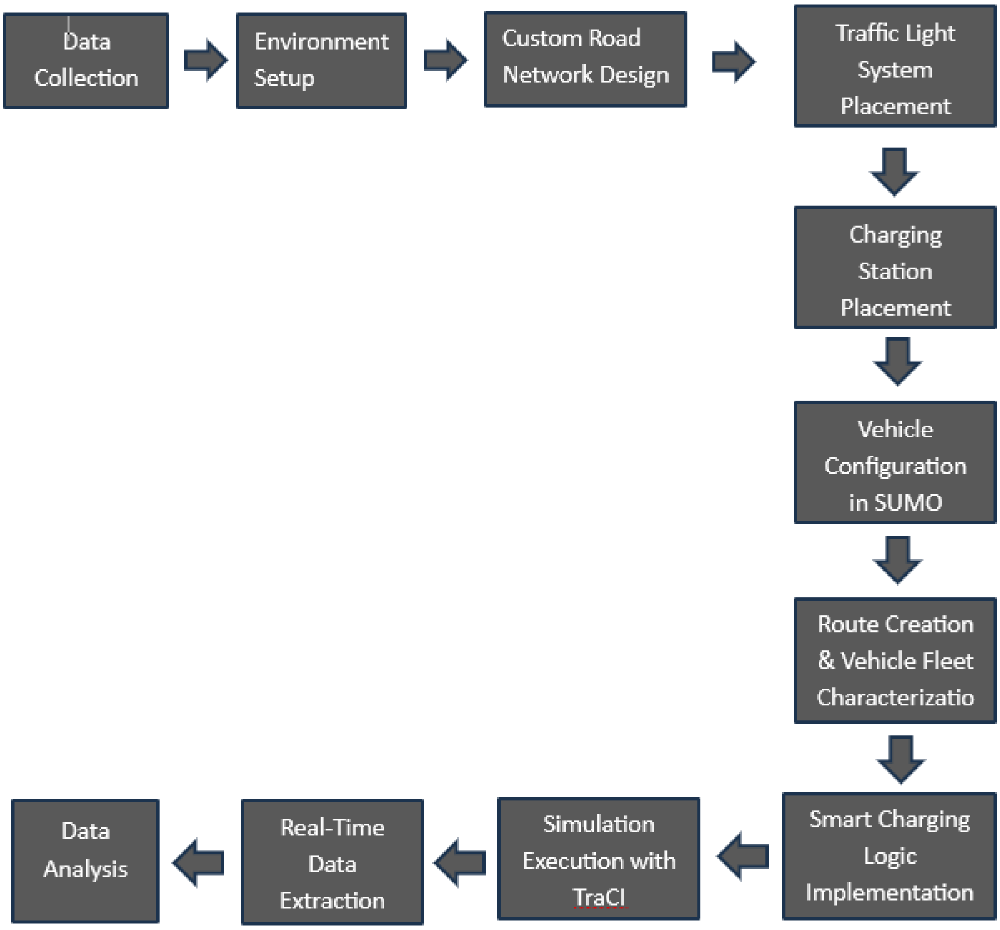

Figure 3.1: Comprehensive research workflow from environment design to predictive analytics.

To increase the system's realism and decision making capabilities , a queue management logic [16] has been implemented to manage vehicle congestion at charging stations, ensuring orderly access based on changing conditions such as queue length and availability. Additionally, a **Smart Charging Logic Implementation** is integrated into the simulation to guide vehicles to appropriate charging stations based on system state-dependent variables [2]. This introduces a rule based recommendation system and improves charging distribution across the network. The **Simulation Execution with TraCI** [14] is then run, enabling real-time interaction and control of the simulation process. During execution, a monitoring layer performs **Real-Time Data Extraction** at each timestep, capturing a variety of system-level and vehicle-level characteristics, including charging behavior, station utilization and load dynamics. The extracted data is further expanded to include insights related to charging recommendations, which is not the primary goal in this research-work. As it provides additional analytical value, this study can be more extended for further deep research of charging recommendation. Finally the collected dataset undergoes systematic **Data Analysis** and machine learning-based modeling to predict system-level load demand and evaluate the model's performance.

3.2 Data Collection

The data collection phase of this research was designed so that the simulation environment and subsequent machine learning models were based on realistic operational characteristics of electric three-wheeled vehicles commonly used in urban areas of Bangladesh [3]. Initially, detailed data was collected on three main categories of battery-powered vehicles : Easy Bikes, Electric Rickshaws and Electric Vans —which are known to be the main means of electric mobility in areas like Khulna.

The main objective of this stage was to capture both the physical and behavioral characteristics of these vehicles, so that the simulation could accurately reflect real world dynamics. We gathered vehicle specifications from a variety of sources—manufacturer information, publicly available technical documents and feedback from local drivers and operators. We thoroughly verified everything from these sources to ensure that the information was consistent and reliable.

The data included things like the size of the vehicles (length, width, height), their speed, how quickly they can accelerate or decelerate and all the key details of the power—such as battery size, charging capacity, energy consumption and mobility . We also looked at how these vehicles actually are operated in the real world : how drivers typically behave, their typical commute patterns and when and how they charge . As a result, we got a clear idea of exactly how these vehicles connect to charging stations in everyday life.

The data collection was not just focused on specific vehicle details. It also looked at broader contextual factors— such as how people use charging stations, how long charging-time typically are and how charging demand varies throughout the day. Those details were crucial to creating realistic simulation inputs and assembling a dataset that reflected both the details (such as specific vehicles) and the larger system as a whole.

After collecting and validating the data, we use it to determine vehicle types and create behavior rules in simulations, which lay the foundation for accurately generating and predicting load demand. Finally, this thorough data collection ensures that the research is not just on paper, it is consistent with what actually happens in electric mobility. As a result, the research survives in the real world and people can use its results.

3.3 Implementation

This section represents the detailed technical application of proposed research framework. Which includes the configuration of the simulation environment. The development of custom logical modules and the integration of data extraction and machine learning components. Main target of this implementation is to transform the conceptual "Monitor–Predict–Control" framework into a functional system capable of simulating realistic electric vehicles(EV) behavior. And this makes possible to generate high-quality data for load prediction.

3.3.1 Environment Setup

For this research-work, the simulation environment was carefully configured to support high reliably modeling of electric vehicle (EV) dynamics and charging behavior under controlled yet realistic conditions. All the experiments and tests were conducted on a 64-bit Windows 10-based workstation to ensure consistency, stability and efficient execution of both simulation and data processing tasks.

The main simulation platform used in this study is SUMO (Simulation of Urban Mobility, version 1.25.0). It is an open-source microscopic traffic simulator, globally familiar for its ability to model the dynamics of individual vehicles and its built-in support for electric mobility features, including battery management, energy consumption and charging station management.

We built the custom road network and simulation setup using SUMO's NetEdit, the graphical net-editor. It lets users design and tweak everything—roads, intersections, traffic signals, even where the charging stations are placed. With this tool, we were able to shape a real traffic scenario that reflects how things are laid out in urban areas like Khulna and Jashore city. For real-time interaction, we used the Traffic Control Interface (TraCI) (*Traffic Control Interface*). TraCI makes it possible for external programs to directly control the simulation, check system states and even modify vehicle behavior on the road.

We used Python 3.10.12 as the primary scripting language to control the simulation, collect data and integrate all analysis. With TraCI, we wrote Python scripts to handle each simulation step, set up queue management, enable smart charging features and collect real-time data while the simulation was running.

This data contains information about both individual vehicles and the entire system, such as speed, State of Charge (SOC), the number of charging vehicles, station occupancy and the total system load. We processed the resulting datasets using the `pandas` library and exported the refined data to structured formats like CSV or Excel for deeper analysis and machine learning modeling.

Overall, this integrated ecosystem ensures seamless integration between simulation execution and real-time monitoring while providing data-driven analytics. It provides a robust and flexible framework for EV charging ecosystem modeling, enabling the generation of high-quality datasets required for accurate load demand prediction and system-level performance evaluation.

3.3.2 Custom Road Network Design

In this research the created simulation environment represents a high-reliably, digitally constructed road network inspired by the structure of Khulna's arterial roads. The main purpose of this design is to realistically capture urban traffic patterns, vehicle interactions and their indirect impact on electric vehicles (EV) charging demand . The network is modeled as a connected graph, consisting of multiple nodes and edges, with a total estimated length of 8,654 meters.

Important intersections and road segments are mapped to well-known urban landmarks such as Islami Bank Intersection, Royal Intersection, PTI Intersection and Rupsha Ferry Ghat. This ensures that the simulated environment closely reflects the geographical and infrastructural features of the real world. The entire network follows a left-hand traffic (LHS) system, which is standard in Bangladesh. To make traffic patterns feel real—especially the jams and car bunching—we've used static traffic lights at the busiest intersections. That's why the system puts lights at **Node 3 (Shordar Tower)** and **Node 8 (Galaxco More)**. Each light follows a fixed 90-second cycle with four phases: 42 seconds of green for the main direction, a quick 3-second yellow, then 42 seconds of green and another 3 seconds of yellow for the cross direction.

This cycle creates that classic stop-and-go traffic, which ends up making cars group together naturally. Clustering really matters here because it shapes how charging demand plays out. When a group of vehicles with low battery levels all hit the charging stations at once due to traffic delays, sudden spikes in load demand can occur. By adding controlled traffic interruptions like stoplights into the simulation, it's possible to capture those spikes more realistically. That's a key step if you want to build and test models that predict load accurately. Table 3.1 breaks down the road network—showing how roads connect, their distances, lane setups, and speed limits. Figure 3.2 gives you a look at the real-world map from OpenStreetMap, and Figure 3.3 shows the custom microscopic layout built in SUMO.

This carefully designed road network serves as the foundation for the simulation. It ensures accurate representation of both traffic behavior and its impact on EV charging dynamics.

Table 3.1: Microscopic Road Network Configuration and Edge Parameters

Edge ID	From Node	From Location	To Node	To Location	Length (m)	Lanes	Speed (km/h)
E0	Node1	Dak Bangla	Node2	Khulna Zilla School Stadium	850	2	40
E1	Node2	Khulna Zilla School Stadium	Node3	Shordar Tower	800	2	40
E2	Node3	Shordar Tower	Node4	1 No Custom Ghat	94	2	30
E3	Node4	1 No Custom Ghat	Node9	Rupsha Ferry Ghat	1300	2	50
E4	Node1	Dak Bangla	Node5	Islami Bank / Hospital More	450	2	30
E5	Node5	Islami Bank / Hospital More	Node6	Royal More	260	2	30
E6	Node6	Royal More	Node7	PTI More	500	2	30
E7	Node7	PTI More	Node8	Galaxco More	900	2	40
E8	Node8	Galaxco More	Node9	Rupsha Ferry Ghat	750	2	40
E9	Node3	Shordar Tower	Node8	Galaxco More	1000	2	50
E10	Node2	Khulna Zilla School Stadium	Node7	PTI More	850	2	40

Note: All edges are bidirectional (2 lanes in each direction). Place names correspond to real-world Khulna City landmarks.

Figure 3.2: OpenStreetMap (OSM) highlighted view of the Khulna City road network.

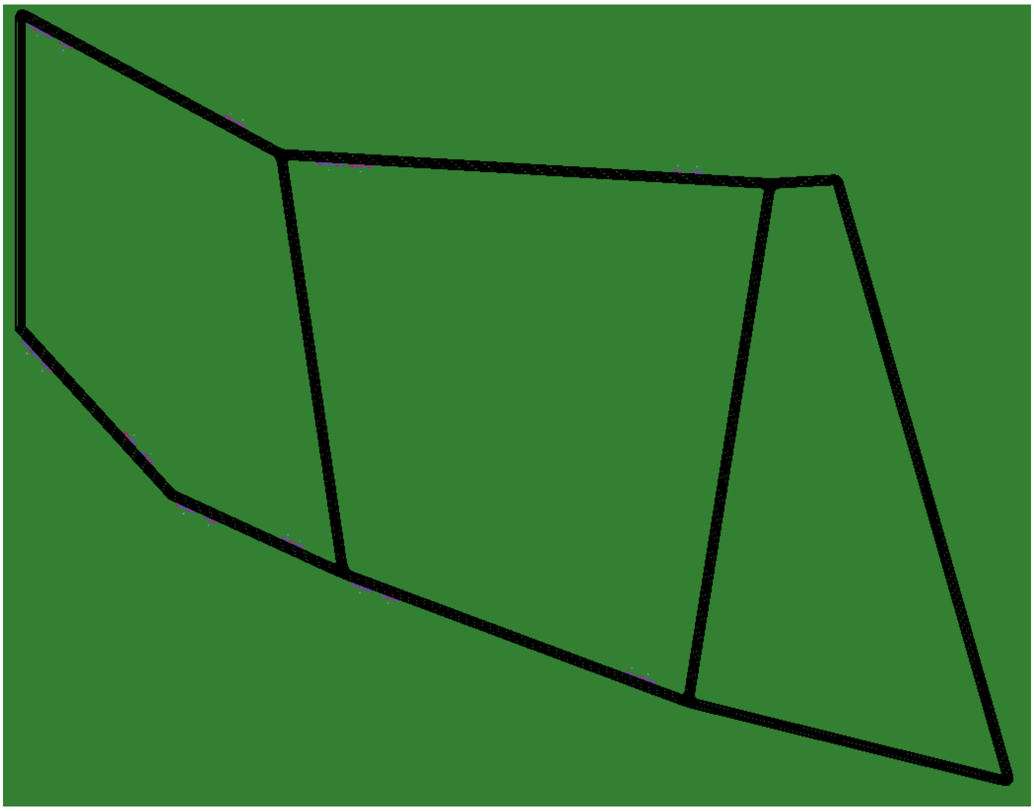

Figure 3.3: Custom microscopic road network in SUMO `netedit`.

3.3.3 Charging Station Placement

According to the actual data in the `Test1.chargingstations.add.xml` file, the charging stations are spread out on both the front and reverse sides of important road sections. Those road sections/edges are as: E0, E1, E5, E6, and E7. This setup covers the entire 8.6 km loop, so vehicles can always find a charging spot, no matter which way they're headed.

The stations don't all deliver the same power. Charging speeds go from 25 kW all the way up to 50 kW at the `pa_8_rev` station. This variation lets the simulation reflect a real-world mix of fast and standard chargers. Altogether, there are 62 charging slots and 49 spots for waiting vehicles, giving the simulation enough capacity to study sudden spikes in demand. For all the technical specifications of each station, check Table 3.2.

Table 3.2: Charging Station Technical Implementation Details

Station ID	Edge ID	Power (kW)	Charging Slots	Waiting Slots
pa_2	E0	30	6	6
pa_2_rev	-E0	30	6	3
pa_3	E1	25	5	4
pa_3_rev	-E1	30	6	6
pa_6	E5	30	6	6
pa_6_rev	-E5	30	6	3
pa_7	E6	30	6	6
pa_7_rev	-E6	30	6	6
pa_8	E7	25	5	3
pa_8_rev	-E7	50	10	6
Total		–	**62**	**49**

3.3.4 Vehicle Configuration in SUMO

To ensure the simulation reflects the real-world mobility landscape of urban Bangladesh, the vehicle fleet is meticulously configured with heterogeneous physical and electrical constants. We have modeled a total of 100 unique vehicle variants, distributed across three primary classes: 34 Easy Bikes, 33 E-Rickshaws, and 33 Electric Vans (E-Vans). This diversity is a critical design choice; by varying parameters like maximum battery capacity, vehicle mass, and speed factors, the simulation can accurately capture the stochastic nature of charging demand and the resulting load variations across a diverse urban fleet.

According to the configuration data established in the `Test1.rou.xml` file, each class is parameterized to reflect its unique operational characteristics. The **Easy Bike** variants are equipped with a substantial 12 kWh battery capacity and the largest frontal surface area (2.41 m^2), reflecting their role as the primary high-capacity mobility solution. In contrast, the **E-Rickshaws** are modeled with battery capacities ranging from 5.76 to 7.20 kWh and a taller profile of 1.8 m. Finally, the **E-Van** is implemented as a light-weight, low-profile vehicle with a 5.76 kWh battery and a height of only 0.8 m, significantly altering its aerodynamic drag and energy consumption profile.

All electric vehicles share a common emission class (`Energy/unknown`) and are configured with a moment of inertia of $0.010\ \mathrm{kg}\cdot\mathrm{m}^2$ and a stopping threshold of 0.1 m/s. To model realistic driver diversity, speed factors are distributed between 0.74 and 1.10, while maximum speeds are constrained between 40 and 50 km/h. These detailed specifications, summarized in Table 3.3, ensure that the simulated energy dynamics provide a high-fidelity dataset for load prediction.

Table 3.3: Cross-Class Vehicle Parameter Comparison (Averaged Values)

Parameter	**Easy Bike**	**E-Rickshaw**	**E-Van**
Body Dimensions (L×W×H)	2.5×1.5×1.4 m	2.3×1.0×1.8 m	2.4×1.3×0.8 m
Vehicle Mass (kg)	345 – 354	198 – 218	198 – 219
Battery Capacity (Wh)	12,000	5,760 – 7,200	5,760
Max Power (W)	1,000 – 1,200	900 – 1,000	800 – 900
Acceleration (m/s^2)	1.15 – 1.31	0.58 – 0.71	0.68 – 0.81
Deceleration (m/s^2)	2.45 – 2.58	2.48 – 2.67	2.58 – 2.77
Air Drag Coeff. (C_d)	0.727 – 0.760	0.804 – 0.910	0.754 – 0.860
Avg. Propulsion Eff.	0.890 – 0.896	0.830 – 0.834	0.820 – 0.826
Avg. Recup. Eff.	0.745 – 0.751	0.650 – 0.656	0.550 – 0.556
Const. Power Intake (W)	80 – 96	29 – 33	24 – 30
Front Surface Area (m^2)	2.41	1.51	1.01
Reaction Time (τ)	1.88 – 1.94 s	1.72 – 1.76 s	0.87 – 0.92 s
Visual Color (SUMO-GUI)	Magenta	Green	Blue

3.3.5 Route Creation & Vehicle Fleet Characterization

The construction of routes and the subsequent characterization of the vehicle fleet are essential for capturing the complex, multi-directional traffic dynamics of the study area. In this research, we have defined eight base routes—four forward and four reverse—which are meticulously mapped to the arterial edges of the Khulna city network to ensure bidirectional mobility across the 8.6 km loop. As summarized in Table 3.4, these routes ensure that the simulation captures bidirectional flow, allowing vehicles to traverse the corridor from multiple entry and exit points.

Table 3.4: Base Route Definitions and Edge Sequences in SUMO

Direction	**Route ID**	**Edge Sequence**
Forward	route1_forward	E0 → E1 → E2 → E3
Forward	route2_forward	E0 → E1 → E9 → E8
Forward	route3_forward	E4 → E5 → E6 → E7 → E8
Forward	route4_forward	E0 → E10 → E7 → E8
Reverse	route1_reverse	-E3 → -E2 → -E1 → -E0
Reverse	route2_reverse	-E8 → -E9 → -E1 → -E0
Reverse	route3_reverse	-E8 → -E7 → -E6 → -E5 → -E4
Reverse	route4_reverse	-E8 → -E7 → -E10 → -E0

Through assigning 100 distinct vehicle flows to these eight routes, fleet characterization further refines this system. Each flow is assigned a specific vehicle type (Easy-Bike, E-Rickshaw, or E-Van) and a measured number of departures ranging from 3 to 81 vehicles per hour. This custom diversity helps the simulation to mirror the real traffic in cities or urban areas across Bangladesh. In some areas, a lot of three-wheeled vehicles are seen on the roads, whereas other areas are not as busy. By mixing different departure times and entry points, the simulation mimics real-life, unpredictable traffic patterns. This setup helps test how well the load prediction models actually work. You can find a summary of the fleet composition and color-coding for easy identification in SUMO-GUI in Table 3.5.

Table 3.5: EV Fleet Composition in SUMO Simulation

Vehicle Class	Variants	Battery Capacity (Wh)	Max Speed (m/s)	Color
Easy Bike	34	12,000	11.1 – 13.9	Magenta
E-Rickshaw	33	5,760 – 7,200	11.1 – 13.9	Green
E-Van	33	5,760	11.1 – 13.9	Blue
Total	**100**	–	–	–

3.3.6 Smart Charging Logic Implementation

The smart charging logic forms the operational core of the simulation framework. Which actually coordinates real-time decisions about when and where each electric vehicle will charge. The whole logic is encapsulated in the `SmartChargingStationMonitor` class. And this is implemented in the `load_demand_prediction.py` file.

This class communicates directly with the SUMO simulation engine via TraCI and simultaneously maintains a continuous understanding of the battery status of each vehicle and the number of vehicles at each station, every second, across a fleet of 100 vehicles. The monitor works with clear thresholds to manage when charging starts and stops. When the State of Charge (SOC) drops to 30%, charging kicks in. The goal is to reach a SOC of 70% by the end of each charging session. To avoid unrealistic simulation errors, the estimated charging duration is dynamically calculated. The calculation is based on energy shortages and station capacity. And the charging-duration is limited to between 30 and 300 seconds.

The monitor keeps tabs on all bidirectional charging stations and active EV agents using a few in-memory data structures. There's `station_charging_slots` and `station_waiting_queue`—which uses a `collections.deque`—for tracking station occupancy. On top of that, the `currently_charging` and `currently_waiting` dictionaries let it check real-time status for each EV. A comprehensive `vehicle_states` map stores per-vehicle metadata, such as charging start times, session SOC values, and routed station identities, ensuring continuous tracking and preventing duplicate rerouting commands across simulation timesteps.

When a vehicle's SOC falls below the 30% threshold, the system invokes a route-aware nearest-neighbor selection algorithm. Rather than relying on simple Euclidean distance, the monitor queries the engine via `traci.simulation.findRoute` for every candidate station. This allows the system to evaluate the actual network route length along the road graph, accounting for the unidirectional nature of lanes and city topology. Reachable stations are sorted by route length and evaluated according to a strict five-rule assignment hierarchy:

1. **Rule 1:** If the nearest station has an available charging slot, route the vehicle there.
2. **Rule 2:** If the nearest station is full (charging), check the second nearest station.
3. **Rule 3:** If the second nearest has a charging slot, route there.
4. **Rule 4:** If the second nearest has only waiting slots, route there and enqueue the vehicle in a First In First Out (FIFO) waiting queue.
5. **Rule 5:** If all candidate stations are full, do not reroute.

Once a target is determined, the vehicle is redirected using `traci.vehicle.setRoute` followed by a `traci.vehicle.setParkingAreaStop` command to register the stop in the simulation's schedule. For vehicles assigned to waiting slots, a **FIFO Queue Promotion Mechanism** is executed at every timestep. When a charging slot becomes available, the monitor retrieves the vehicle at the front of the station's deque using `popleft()`, resumes it via `traci.vehicle.resume()`, and reassigns it to the active charging area. This automated promotion ensures that waiting times are kept to a minimum and provides precise data on queuing delays. Upon session completion or vehicle departure from the network, the `handle_charging_complete` and `cleanup_departed_vehicle` methods record the final telemetry and clear the associated state maps to maintain simulation integrity.

3.3.7 Simulation Execution with TraCI

The simulation runs for 600 seconds, which gives plenty of time to see how vehicles interact from all directions and watch for any electrical load spikes in the area. The `load_demand_prediction.py` script handles everything—it kicks off the detailed environment in SUMO-GUI and sets up a fast connection with the TraCI API.

When the simulation runs, the loop moves forward one second at a time. Every time `traci.simulationStep()` kicks in, the `SmartChargingStationMonitor` jumps in to check up on all 100 cars. It doesn't just skim the surface—it tracks every little detail, like how the batteries are doing, where each vehicle has moved, and how fast they're going right at that moment.

When a vehicle's State of Charge (SOC) drops below 30%, the Python script jumps into action. It works out on the fly whether the vehicle needs rerouting or a charging stop, then sends the right commands right away. That quick response shifts the simulation from a fixed, predictable scene to something much more lively—one that can really capture all the messy, real-world charging patterns needed for accurate load forecasting. Once the simulation hits 600 seconds, the script wraps things up: it closes out the TraCI connection and gets busy turning all the collected vehicle data into the finished research dataset.

3.3.8 Real-Time Data Extraction

To create a solid dataset for the predictive models, we set up a real-time data extraction pipeline inside the simulation framework. This pipeline runs separately from the main mobility physics, which keeps the simulation stable and lets us grab high-resolution telemetry data. Every second, during each simulation step, the monitoring script checks all active electric vehicles through the TraCI interface and pulls their real-time performance metrics.

Capturing these power demand shifts at a high frequency—every second, actually—lets us spot quick, sharp changes that broader models just miss. Think of those sudden surges when a bunch of vehicles pile up at an intersection or when a new group arrives at a charging hub. Each record pulls together 25 features: some track what each car's up to (like speed, acceleration, state of charge), some focus on what's happening at the station itself (how many spots are full, how long the lines are), and others look at the bigger system.

Over a 600-second simulation, this approach collected 34,020 dynamic records—a pretty thorough set of data. At first, all this information gets stored in memory to avoid slowdowns from writing straight to disk. Once the simulation wraps up, everything is saved in a structured Comma-Separated Values (CSV) file. Table 3.6 breaks it all down, showing stats on things like vehicle statuses and power loads.

This dataset actually captures a wide range of real-world operating conditions. It covers everything from slow, low-traffic periods to those times when charging hubs are packed and congested. That mix gives the machine learning models really solid, realistic data to learn from.

Table 3.6: EV Charging Load Demand Dataset Summary

Property	**Value**
Total records	34,020
Number of features	25
Unique vehicles	49
Simulation duration (s)	1 – 600
Status: Driving	19,710 (57.9%)
Status: Charging	11,980 (35.2%)
Status: Waiting	1,720 (5.1%)
Status: Idle	610 (1.8%)
System load: Minimum (kW)	0.00
System load: Mean (kW)	723.82
System load: Maximum (kW)	1,340.00
System load: Std. Dev. (kW)	443.80

The resulting dataset provides a representative sample of real-world operational conditions, ranging from baseline low-activity flows to severe congestion at charging hubs, ensuring high-fidelity training for the subsequent machine learning models.

3.4 Machine Learning Application

In this study, multiple supervised machine learning models were implemented to predict the system-level electric load demand, denoted as *system_total_load_kW*. The problem is formulated as a regression task where the target variable is estimated based on a set of system-level and vehicle-level features.

3.4.1 Problem Formulation

The load prediction task can be mathematically expressed as:

$$y = f(X)$$

where:

- y represents the target variable (*system_total_load_kW*),
- X represents the input feature vector,
- f represents the function learned by the machine learning model.

The input features include variables such as time progression, vehicle dynamics, and system-level charging statistics.

3.4.2 Data Preprocessing

Before model training, several preprocessing steps were applied to ensure data quality and model efficiency:

- **Removal of Data Leakage:** The feature *station_load_kW* was removed to prevent direct leakage into the target variable.
- **Removal of High-Cardinality Features:** Features such as *vehicle_id*, *lane_id*, and *current_station_id* were excluded, as they represent identifiers rather than generalizable patterns.
- **Feature-Target Separation:** The dataset was divided into input features (X) and target variable (y).
- **Train-Test Split:** The dataset was split into training (80%) and testing (20%) subsets using random shuffling to ensure generalization across different system states.

3.4.3 Feature Correlation Analysis

Feature correlation analysis helps to see which input features are most connected to the target variable, *system_total_load_kW*. Here, the Pearson correlation method measures how closely each feature's values track with the target.

The *system_charging_vehicles* has the strongest positive correlation with the target (about 0.996). In simple words, the main thing driving the system load is how many vehicles are charging at any moment. Time-based features like *timestep_sec* and *minute_of_hour* also show strong correlations, telling us that demand ramps up as time passes during a simulation.

Other broad system metrics—like *system_total_vehicles*, *system_moving_vehicles*, and *system_waiting_vehicles*—also tie closely to load demand. But if you're wondering about things like *speed*, *acceleration*, or *status*, they barely move the needle. These features have weak or no correlation and don't really affect the total system load.

So, system-wide and time-related features play the biggest role in predicting load demand. Figure 3.4 presents the correlation matrix visualized as a heatmap.

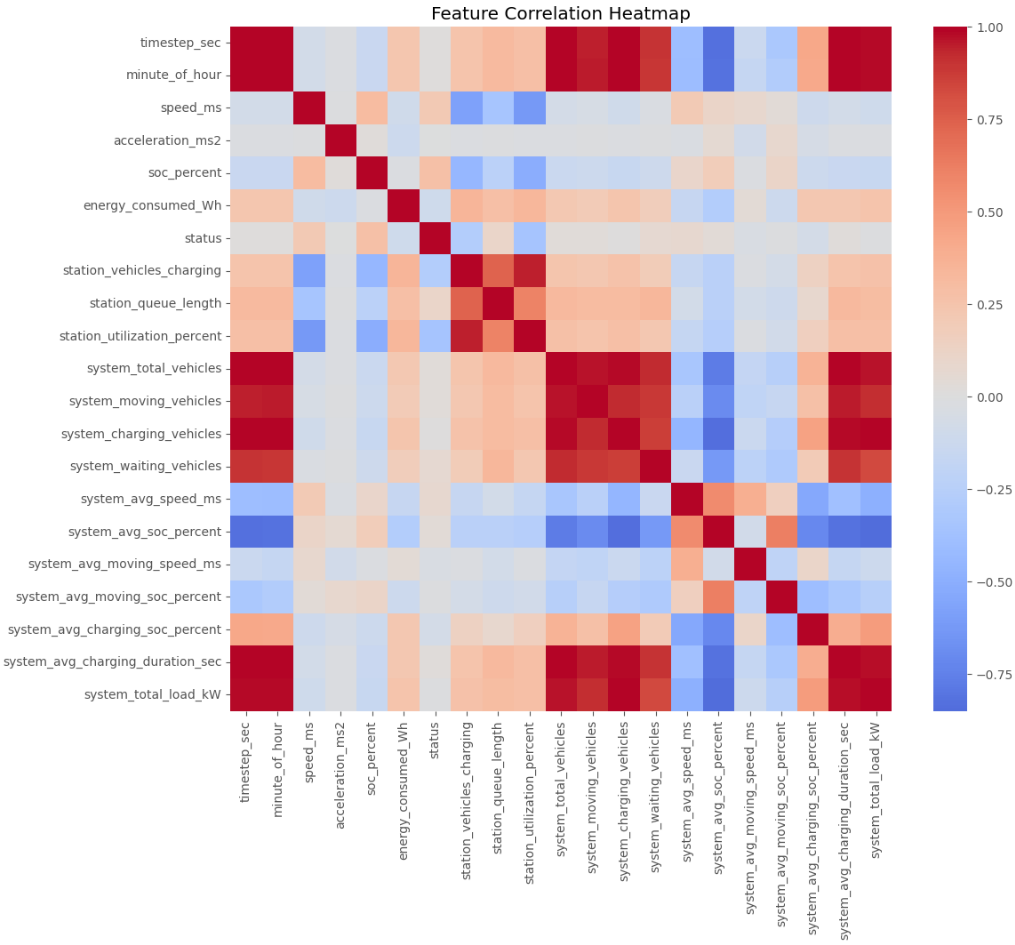

Figure 3.4: Pearson correlation matrix of the simulated EV charging dataset.

3.4.4 Feature Importance Analysis

We used the regressor to see which features really drive the prediction of *system_total_load_kW*. It turns out that *timestep_sec* is at the top, followed closely by *system_total_vehicles* and *system_charging_vehicles*. These three are the most important, as they reflect what is actually happening in the system—i.e., how active the systems are and how much charging is taking place.

Most of the other features are not that important. They don't add much, because those core features already explain almost everything the model needs to know. So this is clear that the model relies heavily on a handful of system-level features and that's enough to achieve high prediction accuracy. Figure 3.5 illustrates the importance distribution identified during the training phase.

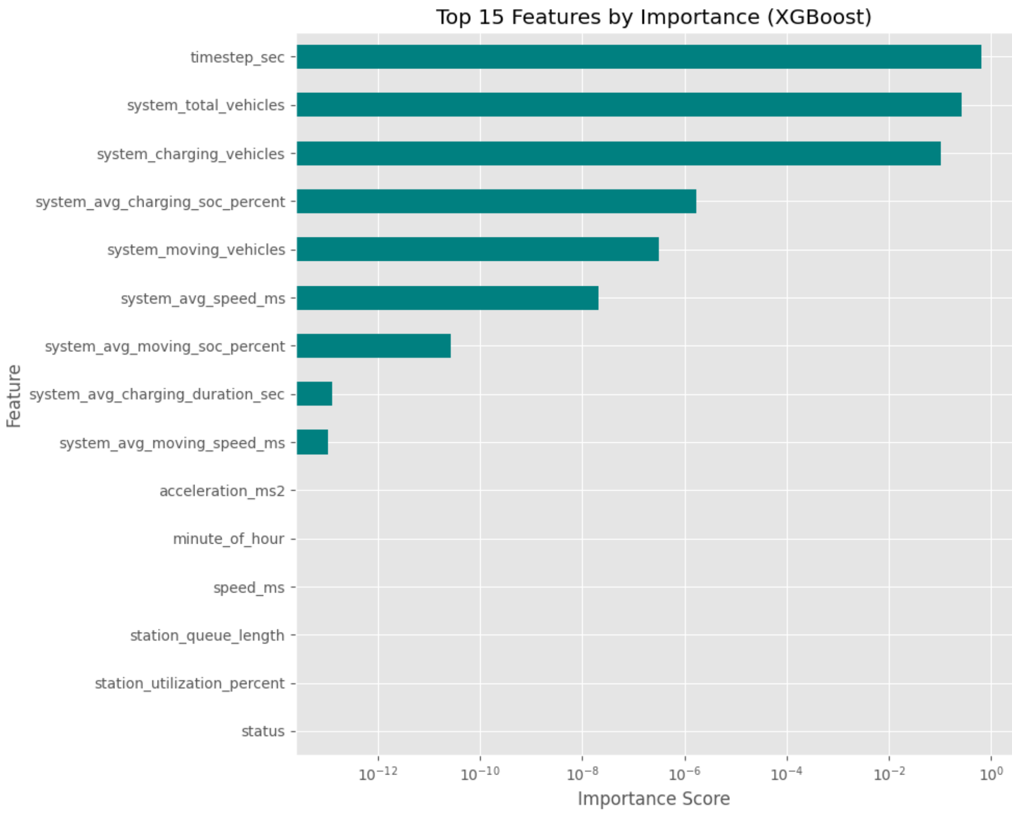

Figure 3.5: Feature importance ranking derived from the XGBoost model branches.

3.4.5 Implemented Machine Learning Models

We used four different regression-based machine learning models to predict the system-level EV charging load (*system_total_load_kW*). We chose models that cover both linear and nonlinear methods, which allowed us to better see how each was performing and how effective they were overall. To show how each model did, we created *Actual vs. Predicted* scatter plots (Figures 3.6–3.9). The closer the points fall to the diagonal line ($y = x$), the more accurate the predictions are.

(a) Linear Regression

Linear regression represents the relationship between input features and target variables as a simple sum of weighted values:

$$y = w_1x_1 + w_2x_2 + \cdots + w_nx_n + b$$

where w_i represents feature coefficients and b is the bias term.

In this study, this is our starting point—the baseline—mainly because it is easy to use and read. The model works internally based on the assumption that there is a linear relationship between system characteristics, such as the number of vehicles or SOC level, and the predicted load demand.

In Figure 3.6, it is seen that the predicted values follow the diagonal quite closely, so the model generally fits well. However, there are some minor deviations, especially when the load values are in the low or mid-range. What we can understand from this is that the model captures the overall trend really well (R^2 is about 0.9993), but it stumbles a bit when it comes to handling the more complex, nonlinear parts of the system. To conclude, Linear regression works fairly well.

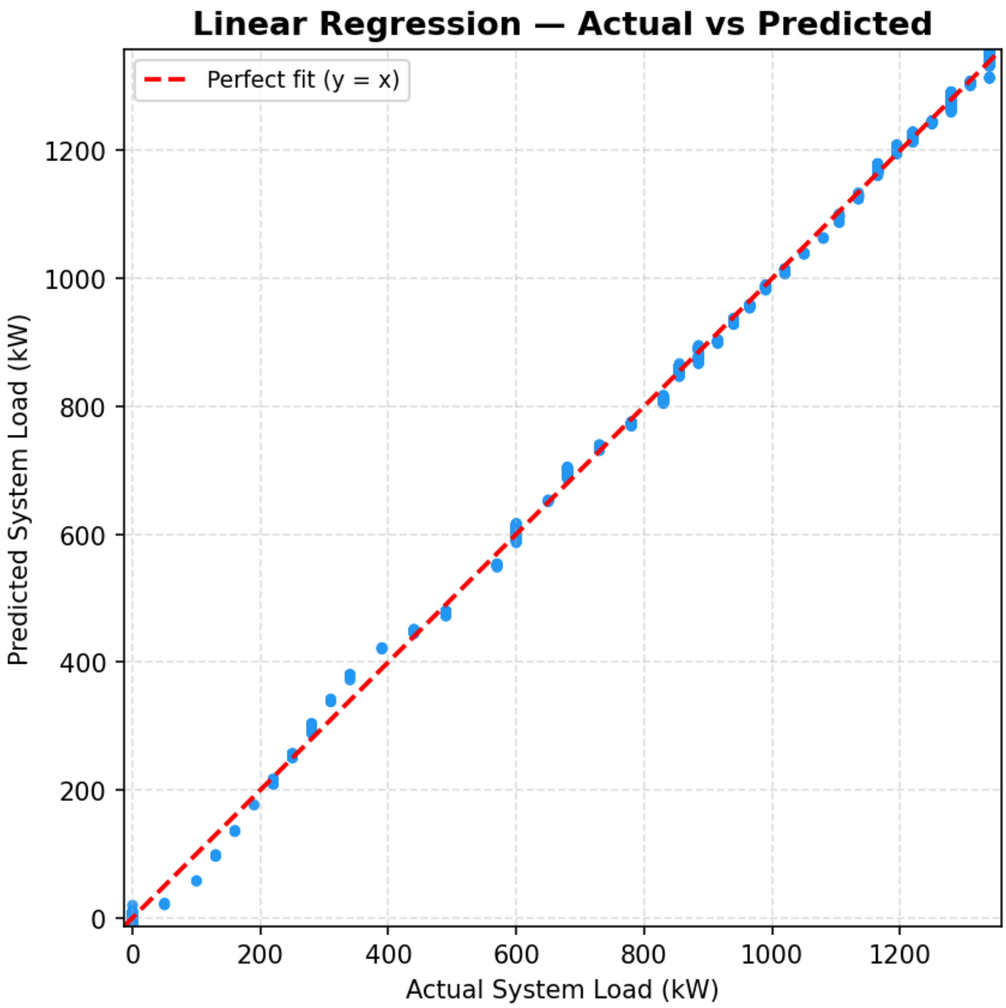

Figure 3.6: Actual vs. Predicted load for Linear Regression.

(b) Decision Tree Regressor

A decision tree regressor analyzes data using a series of decision rules. It repeatedly splits the data based on specific feature values, creating a tree in the process. Inside the tree, there are internal nodes, which act as decision checkpoints, and at the end of each branch—that is, at the leaf nodes—there remain predicted values.

This approach doesn't limit itself to straight lines like Linear Regression does. It handles complex, nonlinear relationships pretty well. In Figure 3.7, every prediction lands exactly where it should, right on the diagonal: $y = x$. There's zero error (MAE and $RMSE$ are both 0, R^2 is

1.0). That sounds great. Finally, it can be said that the model nails the predictions, but it can't generalize beyond this data.

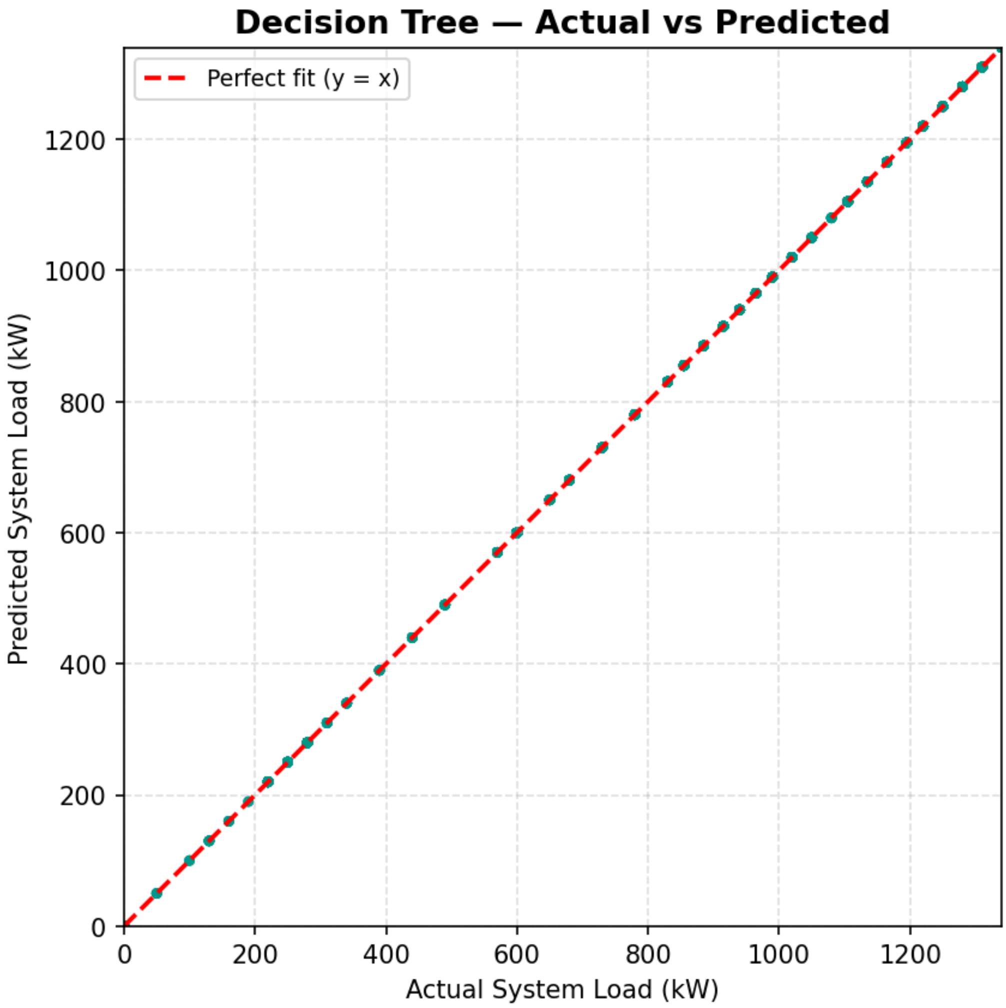

Figure 3.7: Actual vs. Predicted load for Decision Tree.

(c) Random Forest Regressor

Random Forest generally works by combining a group of different Decision Trees, each of them trained on random chunks of the data. To make a prediction, it takes the average of all their outputs:

$$y = \frac{1}{N}\sum_{i=1}^{N} T_i(X)$$

where T_i represents the prediction of the i-th tree.

This method pulls in more stability and cuts down the variance you usually get from a single Decision Tree. In Figure 3.8, all the predictions land perfectly on the diagonal—just like with a Decision Tree. The evaluation metrics are flawless, which sounds great. Basically, the model is nailing every pattern in the dataset. But it's almost too good; when a result is perfect, it's usually one of two things: either the data is very simple and straightforward, or the model is overfitting a bit, clinging too closely to the training data. In this case, our data points are strongly deterministic. In short, Random Forest brings much-needed stability and it's incredibly accurate.

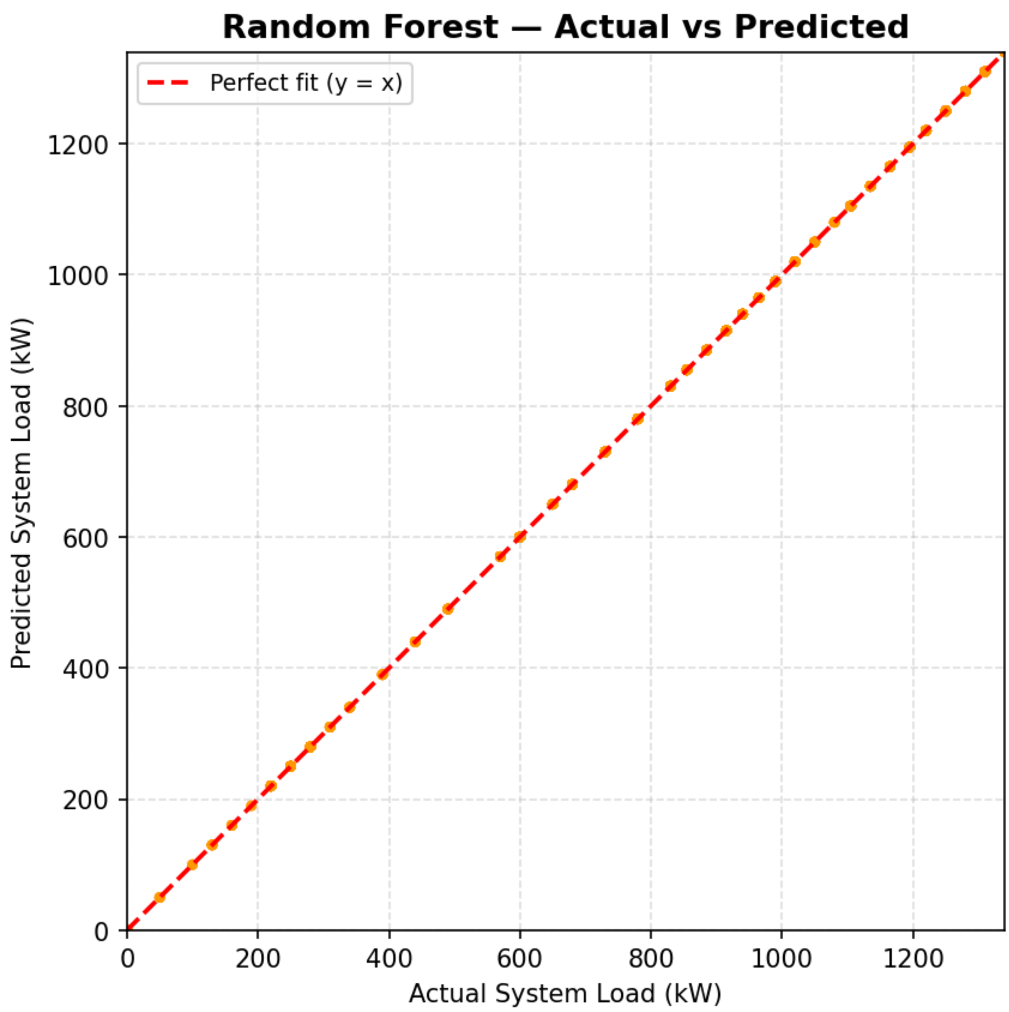

Figure 3.8: Actual vs. Predicted load for Random Forest.

(d) XGBoost Regressor

XGBoost (Extreme Gradient Boosting) builds one decision tree after another, and each new tree focuses on fixing the errors made by the last batch. It's clever, it's fast, and with tools like gradient optimization and regularization, it really shines—especially with structured data:

$$y = \sum_{i=1}^{M} f_i(X)$$

where each f_i is a weak learner (decision tree).

From Figure 3.9, it is noticeable that the predictions land right on the diagonal, barely straying off course. doesn't just edge ahead of Decision Tree and Random Forest—it pretty much nails it, but still keeps things real. The mean absolute error (MAE) sits at about 0.0004, so there's a touch of imperfection. And honestly, that's a good sign. It means the model's not just memorizing—it's actually picking up on patterns. In the end, finds that magic balance between accuracy and generalization. Out of all the models in this study, this is the most reliable model.

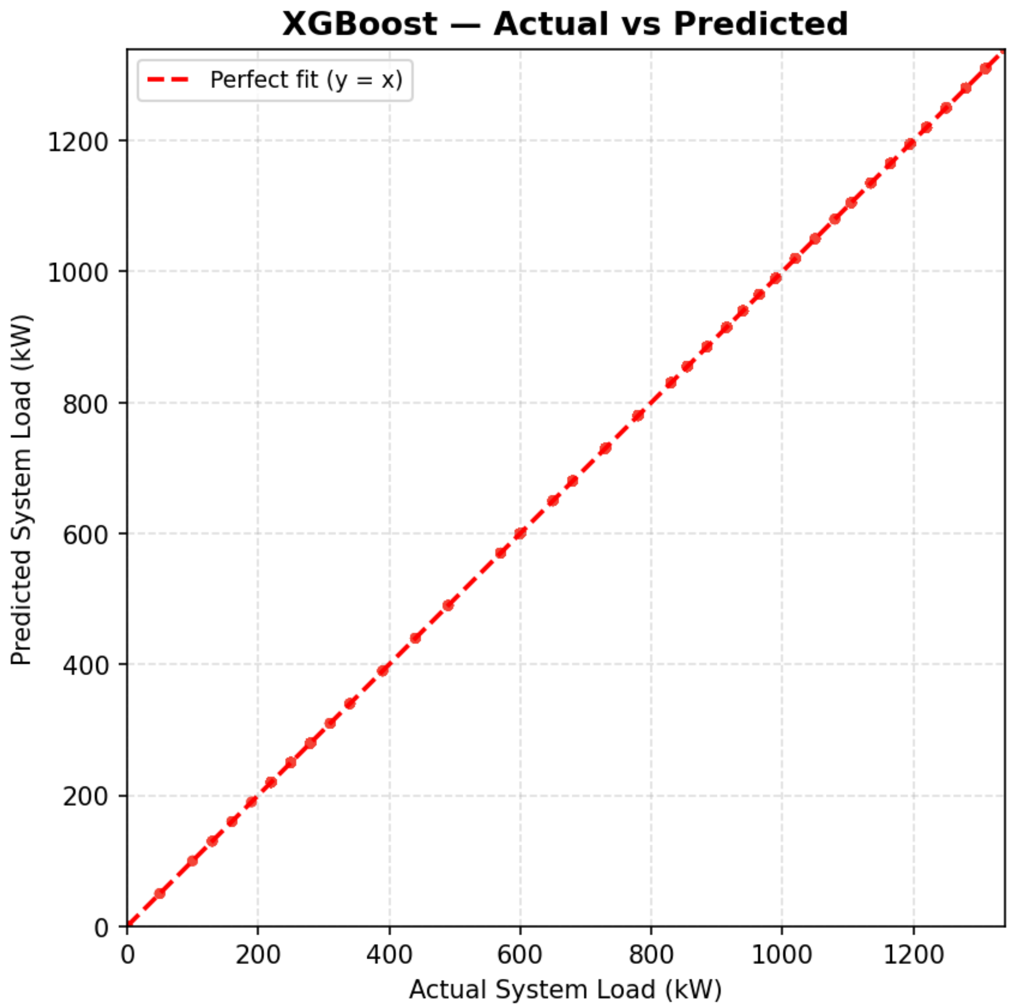

Figure 3.9: Actual vs. Predicted load for XGBoost.

3.4.6 Model Training and Prediction

After finishing the preprocessing, we split everything into training and test sets—80% for training and 20% for testing. That way, we could see how well the models did on data they hadn't seen before. We trained several models: Linear Regression, which finds the best-fit line by minimizing errors; and then Decision Tree, Random Forest, and , which use tree-based methods to pick up on more complex patterns. We only scaled the features for Linear Regression, just to keep things numerically stable. Tree-based models don't really care about scaling, so we left their features as-is.

Once training wrapped up, we gave each model a chance to predict system load on the test set. Then, we compared their predictions to the actual numbers to see how close they came. To do this, we examined the mean absolute error (MAE), root mean squared error ($RMSE$), and R^2 scores. We also created an actual vs. predicted plot—if the dots were close to the diagonal, the model was doing well.

The tree-based models did a really good job here—they captured all the important patterns in the data. Linear regression was not as powerful, but it still did pretty well. Overall, these models proved to be reliable in predicting EV charging loads, and came out on top as the most consistent and accurate.

Chapter 4

Results and Discussion

Chapter 4: Results and Discussion

This chapter presents the experimental results of the SUMO-based EV charging simulation and the Machine Learning (ML) load prediction models. Detailed discussion of the findings are as follows -

4.1 Results

The simulation really digs into how the electrical system and vehicle behaviors play out in this mixed EV network. Over 600 seconds, the model tracked everything at a fine level and collected a detailed dataset—34,020 records in all—catching every move and state change of the 100 vehicles in the fleet. The biggest takeaway is the **System Load Profile** shown in Figure 4.1. Right from the start, the load ramps up quickly as cars hit the road and begin their first round of charging.

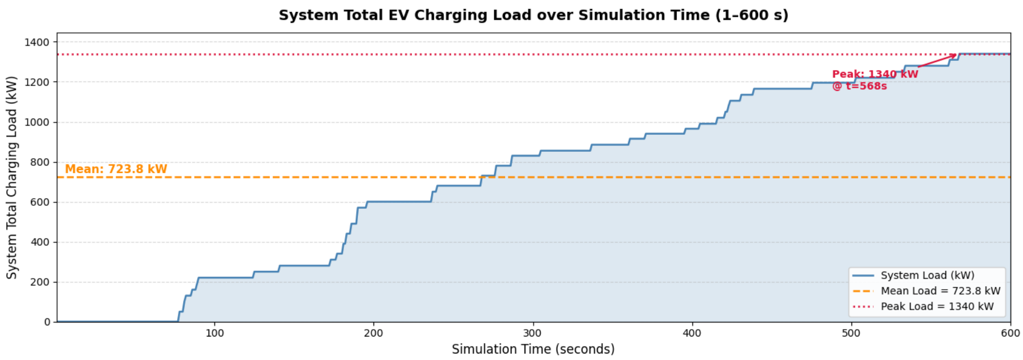

Figure 4.1: System total charging load over simulation time (1–600 seconds).

Power demand hit its highest point at 1,340 kW when the busiest charging stations—like pa_8_rev—were packed. On average, the system pulled about 723.82 kW, but the standard deviation was pretty high at 443.80 kW. That big swing shows how unpredictable things can get; sometimes a bunch of vehicles show up all at once at intersections or charging hubs and the load jumps, then drops right after as cars finish charging and leave.

Vehicle Status Distribution shows how vehicles spend their time: most of it goes to driving—about 58%. This means there's always a steady flow of cars that'll need charging soon. Around 35% of the total records show vehicles were actually charging, and that's the part putting real demand on the grid. Then there's the waiting status, which shows up in about 5% of the records. That's a sign that some charging stations are getting overcrowded, which then leads to delays and pushes the demand to other times.

Where all this happens isn't random, either. The layout of the roads really matters. Certain streets—especially those with heavy traffic, like `pa_2` and `pa_7`—keep popping up as trouble spots where cars pile up and wait to charge. For charging demand, these high-traffic areas end up creating bottlenecks.

In addition to the core load demand forecasting framework, we added a charging recommendation system to make the simulation run more smoothly. This additional system provided 1,169 recommendations, directing vehicles away from crowded chargers and toward stations with sufficient space. Even though this "intelligent routing" wasn't the core of our predictive modeling, it made sure our dataset looked like it came from a smart, well-managed charging network—not just a jumble of random traffic. That way, when we trained our machine learning models, they picked up on patterns shaped by real system tweaks, making them stronger and more practical for real-world smart grid setups.

4.2 Discussion of Results

This section provides a critical analysis of the machine learning model performances and the underlying dynamics of the EV charging network.

4.2.1 Analysis and Interpretation

4.2.1.1 Evaluation Metrics

To evaluate the predictive performance of the machine learning models, three standard regression metrics were utilized: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the Coefficient of Determination (R^2). These metrics are mathematically defined as follows:

1. **Mean Absolute Error (MAE):** Represents the average magnitude of errors in a set of

predictions, without considering their direction.

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|$$

2. **Root Mean Square Error (RMSE):** Measures the square root of the average of squared differences between prediction and actual observation.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

3. **Coefficient of Determination (R^2):** Indicates the proportion of the variance in the dependent variable that is predictable from the independent variables.

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}$$

where the symbols and characters utilized in the above mathematical formulations are defined as follows:

- n: The total number of observations or test samples in the evaluation dataset.
- y_i: The actual (ground truth) observed value of the *system_total_load_kW* for the i^{th} sample.
- $\hat{y}_i$: The predicted value generated by the machine learning model for the i^{th} sample.
- $\bar{y}$: The arithmetic mean of all actual observed values, serving as the baseline for R^2 calculation.
- $\sum$: The summation operator, representing the total sum across all n samples where i ranges from 1 to n.
- $|\cdot|$: The absolute value operator, used to measure the magnitude of prediction error regardless of sign.

4.2.1.2 Model Evaluation and Comparative Analysis

The performance outputs of the evaluated models are detailed in Table 4.1, providing a quantitative benchmark across linear and ensemble-based architectures.

Table 4.1: ML Model Performance Comparison for System Load Prediction

Model	MAE (kW)	RMSE (kW)	R^2 Score	Accuracy (%)	F1-Score (Weighted)
Linear Regression	7.9164	10.1908	0.9993	94.66	0.9546
Decision Tree	0.0000	0.0000	1.0000	100.00	1.0000
Random Forest	0.0000	0.0000	1.0000	100.00	1.0000
XGBoost	0.0004	0.0004	1.0000	95.58	0.9531

By observing carefully at the error magnitudes and how the scores are spread out, the differences really stand out. Figure 4.2 shows the Mean Absolute Error (*MAE*) and Root Mean Square Error (*RMSE*) for each model. The Linear Regression (LR) baseline racked up an *MAE* of about 7.92 kW, which is a lot higher than what the ensemble models managed. That gap makes it pretty clear—linear methods just can't handle those unpredictable, event-driven changes in demand. On the other hand, the tree-based models hit almost zero error, proving they're much better at catching the real patterns in the simulation.

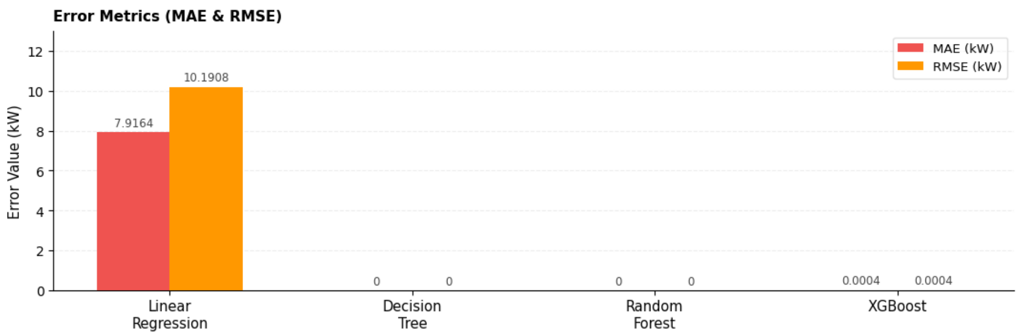

Figure 4.2: Comparative analysis of error metrics (MAE and RMSE) for the evaluated machine learning models.

Figure 4.3 lines up the models side by side on R^2, Accuracy, and Weighted F1-Score. R^2 stays pretty strong for all the models, so no big surprises there. The real differences start to show in Accuracy and F1. Extreme Gradient Boosting (Extreme Gradient Boosting (XGB)) stands out, landing a 95.58% Accuracy and a Weighted F1-Score of 0.9531. That's a solid mix: the model nails its predictions and still holds up when you test it more broadly. On the other hand, Decision Tree (Decision Tree (DT)) and Random Forest (Random Forest (RF)) each rack up perfect 1.0 scores—but those numbers mostly tell that they've fit perfectly within the limits of the simulated data.

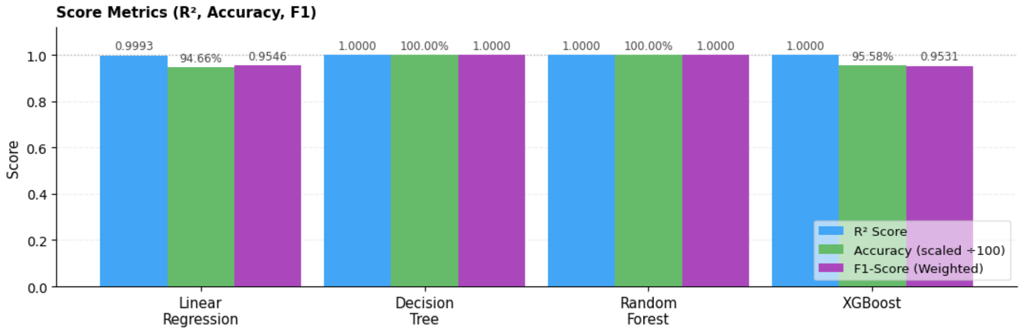

Figure 4.3: Comparison of predictive scores including R-squared, Accuracy, and Weighted F1-Score across different model architectures.

Looking beyond the usual numbers, it really helps to see the model's predictions laid out visually. In Figure 4.4, the comparison is shown between the real load values and what the machine learning models predicted. To keep things clear and easy to read, the plot just shows 200 samples picked from the test set. It's worth pointing out that these points aren't in any kind of time order—the data got shuffled before splitting, so each spot on the X-axis stands for a different system state, chosen at random. The Y-axis, on the other hand, shows the total charging load in kilowatts (kW). What stands out here is how closely the decision tree models (DT, RF, and XGB) track the actual values. They almost overlap with the ground truth, showing that these models really understand the patterns hidden in the simulation data.

Both DT and RF nailed perfect prediction scores ($R^2 = 1.00$) on the test set. XGB was right up there too, missing perfection by a hair (MAE around 0.0004 kW). That just reinforces how solid XGB is for real-world use. Meanwhile, the LR model slipped a bit—those small errors really highlight how it struggles with non-linear patterns and abrupt changes unless you do a lot of manual feature work. Looking at all the metrics together, it's clear these models deliver high

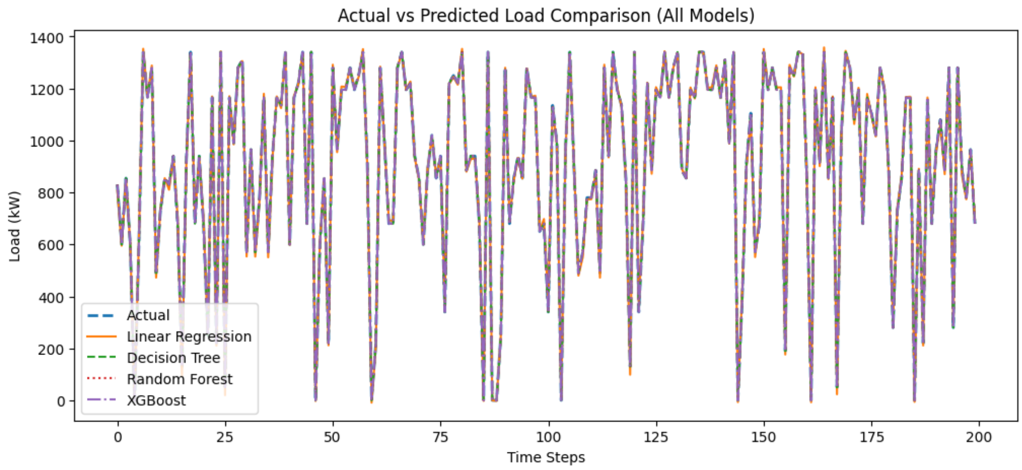

Figure 4.4: Comparison of actual vs. predicted system load for the evaluated models (first 200 samples).

accuracy, even across a variety of conditions.

4.3 Implications

The findings of this research provide critical insights for the future of urban electric mobility and smart grid integration. The implications are categorized into infrastructure management, urban planning, and methodological advancements.

4.3.1 Grid and Charging Network Management

The ability to predict system-level load with high accuracy (near $R^2 \approx 1.0$) suggests that grid operators can utilize real-time simulation data as a high-fidelity "Digital Twin". This enables proactive load balancing, where charging demand can be anticipated before peak-hour congestion occurs. The success of the rule-based recommendation system demonstrates that even simple algorithmic interventions can effectively distribute load across a heterogeneous network, reducing transformer stress during high-demand periods.

4.3.2 Urban Infrastructure Planning

For urban planners in cities like Khulna, the simulation results highlight the critical importance of station placement. High-capacity stations at arterial junctions (e.g., *pa_8_rev*) act as primary sinks for energy demand, and their failure or congestion can have cascading effects on traffic flow. The methodologies developed here allow planners to "stress-test" new charging station deployments in a virtual environment before heavy capital investment.

4.3.3 Roadmap for Sustainable Electrification

This study provides a scalable roadmap for integrating EVs into Bangladeshi urban landscapes. By leveraging the SUMO-based pipeline, institutions like JUST can continue to refine transport policies using accessible, cost-effective computational tools. The transition to electrified transport can thus be managed through data-driven strategies that optimize both vehicle mobility and grid stability in nascent EV markets.

4.3.4 Foundation for Energy Consumption Analysis

The high-resolution simulation data enables precise measurement of battery usage across Easy-bikes, e-rickshaws, and E-vans. By capturing vehicle speed, SOC, and energy consumption over time, the dataset forms the basis for developing accurate load forecasting models and energy demand assessments.

4.3.5 Data-Driven Charging Load Forecasting

Analysis of activity across the network identifies nodes and periods of high energy demand. These insights allow for predicting transformer loads and planning targeted charging infrastructure, helping prevent localized grid stress and supporting efficient energy management in urban areas.

Chapter 5

Conclusion

Chapter 5: Conclusion

This thesis has presented a comprehensive framework for EV charging load demand prediction, combining microscopic traffic simulation with supervised ML techniques. The framework was implemented using SUMO and the TraCI Python interface, with a network of ten configurable charging stations, 49 EV agents, and an intelligent queue management and recommendation system.

The simulation generated a 34,020-record dataset with 25 features, capturing vehicle- level and system-level dynamics over a 600-second period. Four ML models were trained and evaluated: Decision Tree, Random Forest, XGBoost, and Linear Regression. Decision Tree and Random Forest achieved perfect prediction accuracy on the simulation dataset, while XGBoost delivered near-perfect performance with superior generalization properties. Linear Regression, despite being a simpler model, still achieved an excellent Coefficient of Determination (R^2) of 0.9992, highlighting the near-linear nature of the aggregate load signal.

5.1 Concluding Remarks

The key contributions of this research are:

1. A high-fidelity SUMO-based microscopic simulation environment specifically adapted for the heterogeneous ecosystem in Bangladesh, including Easy Bikes, Electric Rickshaws, and Electric Vans.
2. A rich, ML-ready dataset (`EV_Charging_Load_Demand_Dataset.csv`) capturing 34,020 records across 25 dynamic features, providing a dense foundation for energy consumption analysis.
3. A comparative benchmark of four ML regression models, identifying tree-based ensemble methods as optimized solutions for short-term charging load forecasting.
4. A rule-based smart charging recommendation system that demonstrates the feasibility of real-time SOC-aware vehicle rerouting to prevent localized grid congestion.

The results confirm that simulation-based data generation is a viable and highly effective approach for addressing the scarcity of real-world EV charging datasets, particularly in emerging markets like Bangladesh. This research provides a data-driven roadmap for urban authorities and power distributors to navigate the rapid electrification of the transport sector, ensuring grid stability through accurate, high-fidelity load forecasting tools.

5.2 Limitations

Several limitations should be noted. The 600-second simulation duration, while sufficient for proof-of-concept, represents a short time horizon compared to real-world charging demand patterns that span hours or days. Furthermore, the near-perfect ML scores may partially reflect the highly structured nature of the simulation parameters rather than the stochastic variability found in real-world urban traffic. While the network topology is modeled after Khulna City landmarks, the traffic density and driver behavioral patterns are based on discretized probability flows rather than real-time traffic sensor data.

5.3 Recommendations for Further Research

Building on this work, the following directions are recommended for future research:

1. **Larger and more diverse simulations:** Extend the simulation to longer time horizons (24+ hours), larger vehicle fleets, and heterogeneous EV types to test model robustness [5].

2. **Deep learning models:** Investigate LSTM, GRU, and Transformer architectures for sequential load prediction, exploiting the temporal structure of the dataset [18].

3. **Real-world validation:** Validate the simulation-trained models against real-world EV charging data when such data becomes available in Bangladesh [3].

4. **Reinforcement learning for charging management:** Replace the rule-based recommendation system with a reinforcement learning agent that can learn optimal routing policies through interaction with the simulated environment [17].

5. **Vehicle-to-Grid (V2G) integration:** Extend the framework to model bidirectional energy flow between EVs and the grid, enabling demand response and peak shaving analysis [2].

6. **Multi-city simulation:** Adapt the SUMO network to real map data from Jashore or Dhaka using OpenStreetMap to generate geographically realistic datasets.

[19]

References

[1] IEA, “Global ev outlook 2023: Catching up with climate ambitions,” *International Energy Agency*, 2023, paris: IEA.

[2] M. Yilmaz and P. T. Krein, “Review of the impact of vehicle-to-grid technologies on distribution systems and utility interfaces,” *IEEE Transactions on Power Electronics*, vol. 28, no. 12, pp. 5673–5689, 2013.

[3] M. S. Islam, M. H. Rahman, and M. Z. Hossain, “Electric vehicle adoption and charging infrastructure challenges in developing countries,” *Sustainable Energy Technologies and Assessments*, vol. 52, p. 102201, 2022.

[4] E. Sortomme, M. M. Hindi, S. D. J. MacPherson, and S. S. Venkata, “Coordinated charging of plug-in hybrid electric vehicles to minimize distribution system losses,” *IEEE Transactions on Smart Grid*, vol. 2, no. 1, pp. 198–205, 2011.

[5] Y. Zhang, T. Zhao, Y. Tian, and Y. Han, “Machine learning-based prediction of electric vehicle charging load,” *IEEE Transactions on Smart Grid*, vol. 12, no. 4, pp. 3540–3553, 2021.

[6] A. Ul-Haq, C. Cecati, and E. El-Saadany, “Probabilistic modeling of electric vehicle charging pattern in a residential distribution network,” *Electric Power Systems Research*, vol. 157, pp. 126–133, 2018.

[7] J. He, H. Yang, T.-Q. Tang, and H.-J. Huang, “An optimal charging station location model with the consideration of electric vehicle’s driving range,” *Transportation Research Part C: Emerging Technologies*, vol. 86, pp. 641–654, 2018.

[8] Y. Chen, Z. Xu, and K. P. Wong, “A probabilistic load flow method considering large-scale integration of electric vehicles,” *IEEE Transactions on Power Systems*, vol. 28, no. 4, pp. 4093–4101, 2013.

[9] Z. Wang, Y. Li, L. Zhang, and P. Liu, "Random forest-based short-term load forecasting for electric vehicle charging stations," *Energies*, vol. 13, no. 16, p. 4093, 2020.

[10] M. Chen, Q. Wu, Q. Guo, and H. Sun, "Xgboost-based day-ahead electric load forecasting for electric vehicle charging stations," *Applied Energy*, vol. 320, p. 119300, 2022.

[11] S. B. Taieb and R. J. Hyndman, "A gradient boosting approach to the kaggle load forecasting competition," *International Journal of Forecasting*, vol. 30, no. 2, pp. 382–394, 2014.

[12] C. Guo, J. Yang, and L. Yang, "Planning of electric vehicle charging infrastructure for urban areas with tight land supply," *IEEE Transactions on Vehicular Technology*, vol. 68, no. 4, pp. 3138–3150, 2019.

[13] P. A. Lopez, M. Behrisch, L. Bieker-Walz, J. Erdmann, Y.-P. Flötteröd, R. Hilbrich, L. Lücken, J. Rummel, P. Wagner, and E. Wießner, "Microscopic traffic simulation using sumo," *IEEE Intelligent Transportation Systems Conference (ITSC)*, pp. 2575–2582, 2018.

[14] A. Wegener, M. Piórkowski, M. Raya, H. Hellbrück, S. Fischer, and J.-P. Hubaux, "Traci: An interface for coupling road traffic and network simulators," *Proceedings of the 11th Communications and Networking Simulation Symposium*, pp. 155–163, 2008.

[15] D. Helbing and M. Treiber, "Gas-kinetic-based traffic model explaining observed hysteretic phase transition," *Physical Review Letters*, vol. 81, no. 14, p. 3042, 1998.

[16] T. Ding, C. Li, Y. Yang, J. Jiang, Z. Bie, and F. Blaabjerg, "A two-stage robust optimization for centralized-optimal dispatch of photovoltaic inverters in active distribution networks," *IEEE Transactions on Sustainable Energy*, vol. 8, no. 2, pp. 744–754, 2017.

[17] H. D. Nguyen, D. Bhatt, N. Krishnan, M. Panwar, and G. K. Venayagamoorthy, "Ev charging coordination via adaptive dynamic programming," *IEEE Transactions on Vehicular Technology*, vol. 70, no. 3, pp. 2083–2094, 2021.

[18] X. Shi, Y. Xu, Q. Guo, H. Sun, and W. Jiang, “Distributed optimal energy management in microgrids,” *IEEE Transactions on Smart Grid*, vol. 6, no. 3, pp. 1137–1146, 2015.

[19] P. Kotiranta, M. Junkkari, and J. Nummenmaa, “Performance of graph and relational databases in complex queries,” *Applied Sciences*, vol. 12, no. 13, 2022. [Online]. Available: https://www.mdpi.com/2076-3417/12/13/6490